

Russian Language Toxic Comments

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Abstract

This paper investigates the task of automatic toxic comment classification in the Russian language using natural language processing techniques. A TF-IDF representation combined with a Logistic Regression classifier was implemented as a baseline approach. To improve classification performance, transformer-based models were explored, including ruBERT-tiny and ruBERT-base fine-tuned using the parameter-efficient LoRA (Low-Rank Adaptation) method. The workflow included text preprocessing, tokenization, model training, and evaluation using Accuracy, Precision, Recall, and F1-score metrics. The experimental results demonstrate the effectiveness of transformer-based approaches over traditional text representation methods and confirm the effectiveness of modern language models for toxic content detection tasks. Project code is available at: <https://github.com/Sapushiro/russian-toxic-comments-classification>.

1 Introduction

With the rapid growth of social media platforms, online forums, and user-generated content, the amount of textual information available on the Internet continues to increase. At the same time, the presence of toxic comments, including insults, harassment, hate speech, and offensive language, has become a significant challenge for online communities. Toxic content negatively affects user experience, encourages conflicts, and may create unsafe communication environments. Therefore, the automatic detection of toxic comments has become an important task in the field of Natural Language Processing (NLP).

Traditional approaches to toxic comment classification often rely on hand-crafted text representations such as TF-IDF combined with machine learning classifiers. Although these methods are computationally efficient, they are limited in their ability to capture contextual information and semantic relationships between words. Recent advances in transformer-based language models, particularly BERT-based architectures, have significantly improved performance in various text classification tasks by learning contextual representations of language.

The objective of this work is to investigate different approaches to Russian toxic comment classification and compare their effectiveness. Both traditional

machine learning methods and modern transformer-based models are considered. A TF-IDF representation with Logistic Regression is used as a baseline model, while ruBERT-tiny and ruBERT-base models are explored as advanced transformer-based solutions for binary toxicity classification.

The main contribution of this work is a comparative analysis of classical and transformer-based approaches on the same dataset, as well as the application of parameter-efficient fine-tuning using LoRA (Low-Rank Adaptation). This technique enables the adaptation of large language models while significantly reducing the number of trainable parameters and computational requirements. The obtained results provide insights into the effectiveness of modern language models for toxic content detection in Russian texts.

1.1 Team

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2 Related Work

The task of automatically detecting toxic comments is one of the most demanded tasks of natural language processing. For the Russian language, one of the most well-known open datasets is the Russian Language Toxic Comments, published on the Kaggle platform [1]. Due to the openness of the dataset, a large number of solutions have been implemented based on it, from classical machine learning methods to modern transformer models.

One of the most famous studies is the work of VK "Toxic Comments Detection in Russian" [2]. The authors conducted an additional quality check of the dataset markup and compared several approaches for classifying toxic comments. The study examined the Multinomial Naive Bayes, BiLSTM, multilingual BERT, ruBERT, as well as two versions of the Multilingual Universal Sentence Encoder. The best result was demonstrated by the ruBERT-Toxic model, achieving Precision = 91.91%, Recall = 92.51% and F1-score = 92.20% [2]. The authors have shown that transformer models are significantly superior to classical machine learning methods and recurrent neural networks due to a better contextual representation of the text.

In addition to the original VK research, Kaggle presents practical implementations of various approaches to solve this problem. One of the most interesting solutions is to use the DeepPavlov library and the pre-trained RuBERT model to classify toxic comments [3]. The work uses the transformer architecture of BERT, adapted for the Russian language, which allows you to take into account the context of words and the semantic connections between them. The authors note the high quality of the classification and demonstrate the advantages of transformer models compared to classical approaches based on a bag of words and statistical features.

Other published solutions use traditional text processing methods [4]. The most common approach is to represent comments using TF-IDF vectorization

followed by training classifiers such as Logistic Regression, Random Forest, and Naive Bayes. Such methods have low computational requirements and a high learning rate, but they are usually inferior to transformer-based models in terms of classification quality, especially in cases of complex language constructions, sarcasm, or implicit toxicity [4].

In recent years, special attention has been paid to parametrically efficient methods of retraining large language models. One such approach is Low-Rank Adaptation (LoRA), which allows to adapt a pre-trained model by training a small number of additional parameters without changing the initial weights of the model [5]. Using LoRA significantly reduces the requirements for computing resources and memory, while maintaining quality close to full fine-tuning [5].

This paper considers a comparison of several approaches to classifying toxic comments in Russian: the classical TF-IDF + Logistic Regression model, the compact transformer ruBERT-tiny model, and the ruBERT-base model, which was further trained using LoRA. This set of models makes it possible to assess the trade-off between computational efficiency and classification quality, as well as compare modern methods with traditional machine learning approaches.

3 Model Description

The proposed solution addresses the task of binary toxic comment classification in Russian. The overall workflow consists of data preprocessing, dataset splitting, model training, and performance evaluation. Three different approaches were investigated: a classical machine learning baseline based on TF-IDF and Logistic Regression, a compact transformer model ruBERT-tiny, and a larger ruBERT-base model fine-tuned using the Low-Rank Adaptation (LoRA) technique.

The dataset contains Russian-language comments labeled as either toxic or non-toxic. Before training, the data were divided into training, validation, and test subsets. The validation set was used for hyperparameter tuning and model selection, while the test set was reserved for the final evaluation.

The first approach uses the TF-IDF (Term Frequency–Inverse Document Frequency) representation. TF-IDF converts each document into a numerical vector according to

$$TF - IDF(t, d) = TF(t, d) \cdot \log \left(\frac{N}{df(t)} \right), \quad (1)$$

where $TF(t, d)$ denotes the frequency of term t in document d , N is the total number of documents, and $df(t)$ is the number of documents containing term t .

The resulting TF-IDF vectors were used as input features for the Logistic Regression classifier. Logistic Regression estimates the probability that a comment belongs to the toxic class using the sigmoid function. This approach serves as a baseline model for comparison with transformer-based architectures.

The second approach is based on the ruBERT-tiny transformer model. This model is a compact version of ruBERT containing significantly fewer parameters

than the original architecture. Text comments were tokenized using the corresponding tokenizer and truncated to a maximum sequence length of 128 tokens. The transformer encoder generates contextualized token representations, and the representation of the special classification token is passed to a classification layer that predicts the probability of toxicity.

The third approach employs the ruBERT-base model with Low-Rank Adaptation (LoRA). Instead of updating all model parameters during fine-tuning, LoRA introduces trainable low-rank matrices into selected transformer layers. The adapted weight matrix is represented as

$$W' = W + BA, \quad (2)$$

where W is the original frozen weight matrix and A and B are trainable low-rank matrices. This approach significantly reduces the number of trainable parameters and lowers memory consumption while preserving most of the performance of the original model.

AdaLoRA is an extension of LoRA that dynamically allocates low-rank adaptation parameters during training. Instead of using a fixed rank throughout the optimization process, AdaLoRA adjusts the rank according to parameter importance, enabling more efficient utilization of the adaptation budget.

Model performance was evaluated using Accuracy, Precision, Recall, and F1-score. Precision measures the proportion of correctly predicted toxic comments among all comments predicted as toxic, while Recall measures the proportion of correctly identified toxic comments among all actual toxic comments.

The primary evaluation metric used in this work is the F1-score, defined as

$$F_1 = \frac{2 \cdot \textit{Precision} \cdot \textit{Recall}}{\textit{Precision} + \textit{Recall}}. \quad (3)$$

The F1-score provides a balanced assessment of classification quality and is particularly suitable for binary classification problems with potentially imbalanced class distributions.

4 Dataset

The experiments were conducted on the Russian Language Toxic Comments dataset available on the Kaggle platform. The dataset contains user-generated comments written in Russian and annotated for binary toxicity classification. Each comment belongs to one of two classes: toxic or non-toxic. The objective of the task is to automatically determine whether a comment contains toxic content.

The dataset consists of 14,412 comments, including 4,826 toxic comments and 9,586 non-toxic comments. Therefore, the dataset is moderately imbalanced, with non-toxic comments occurring approximately twice as often as toxic comments. For this reason, model performance was evaluated using not only Accuracy but also Precision, Recall, and F1-score.

To ensure a fair evaluation of the proposed approaches, a stratified train-validation-test split was applied while preserving the original class distribution. The dataset was divided into training (80%), validation (10%), and test (10%) subsets. The resulting statistics are presented in Table 1.

Table 1: Dataset statistics	
Property	Value
Total comments	14,412
Toxic comments	4,826
Non-toxic comments	9,586
Training samples	11,528
Validation samples	1,442
Test samples	1,442

The training subset was used for model fitting, the validation subset for hyperparameter tuning and model selection, and the test subset for the final evaluation. Stratified sampling ensured that the ratio of toxic and non-toxic comments remained approximately constant across all subsets, reducing the risk of biased evaluation results.

5 Experiments

5.1 Metrics

The performance of all models was evaluated using Accuracy, Precision, Recall, and F1-score. Accuracy measures the overall proportion of correctly classified comments. Precision measures the proportion of correctly predicted toxic comments among all comments classified as toxic, while Recall measures the proportion of actual toxic comments that were successfully identified by the model.

The primary evaluation metric used in this work is the F1-score, which combines Precision and Recall into a single measure:

$$F_1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}. \quad (4)$$

Since the dataset is moderately imbalanced, the F1-score provides a more informative evaluation than Accuracy alone. In addition to class-specific metrics, Macro F1-score and Weighted F1-score were also analyzed to assess overall model performance.

5.2 Experiment Setup

For transformer-based experiments, two architectures were evaluated: ruBERT-tiny and DeepPavlov ruBERT-base-cased. The ruBERT-base model was used

as the backbone architecture for LoRA and AdaLoRA parameter-efficient fine-tuning experiments.

The dataset was divided into training, validation, and test subsets using stratified sampling. Texts were tokenized using the original ruBERT tokenizer and truncated to a maximum sequence length of 256 tokens. Dynamic padding was applied during batch preparation. The random seed was fixed to ensure reproducibility.

For LoRA experiments, low-rank adaptation matrices were injected into the self-attention and feed-forward layers of the transformer. The rank parameter was set to $r = 32$, the scaling coefficient was $\alpha = 64$, and the dropout rate was 0.05. Adaptation modules were applied to the **query**, **key**, **value**, and **dense** layers of all transformer blocks. The classifier head was additionally preserved.

The resulting LoRA model contained 5,359,106 trainable parameters out of 183,214,084 total parameters, corresponding to only 2.93 % of the full model.

For AdaLoRA experiments, the initial rank was set to 32 and dynamically reduced to a target rank of 16 during training. The adaptive rank allocation process used $\beta_1 = 0.85$, $\beta_2 = 0.85$, initialization step $t_{init} = 200$, final adaptation step $t_{final} = 1000$, and update interval $\Delta T = 10$. The total number of optimization steps was 2884. The same adaptation targets (**query**, **key**, **value**, **dense**) and dropout rate of 0.05 were used as in the LoRA configuration. The AdaLoRA model contained 10,719,010 trainable parameters out of 188,574,061 total parameters, corresponding to 5.68% of all model parameters.

5.3 Baselines

As a baseline approach, TF-IDF vectorization combined with Logistic Regression was used. This method is widely employed in text classification tasks due to its simplicity, efficiency, and strong performance on sparse textual features.

The TF-IDF representation converts each comment into a numerical feature vector based on term frequency and inverse document frequency statistics. These vectors are then used to train a Logistic Regression classifier that predicts whether a comment is toxic or non-toxic.

The baseline model serves as a reference point for evaluating the benefits of transformer-based architectures and parameter-efficient fine-tuning techniques.

6 Results

Table 2 presents the performance of all evaluated approaches on the Russian toxic comment classification task. The comparison includes classical machine learning methods, transformer-based models, parameter-efficient fine-tuning approaches, and previously published results from the literature.

The TF-IDF + Logistic Regression baseline achieved a Toxic F1 score of 0.78. While this result demonstrates that traditional machine learning methods can provide a reasonable baseline, their performance is noticeably lower than that of transformer-based architectures. This observation is consistent with previous

Table 2: Comparison of toxic comment classification approaches (* models trained in this work)

Model	Toxic F1	Macro F1
TF-IDF + Logistic Regression	0.78	0.84
DeepPavlov Fine-Tuning (Kaggle)	0.83	0.88
*DeepPavlov ruBERT + LoRA	0.87	0.90
*ruBERT-tiny	0.88	0.91
*DeepPavlov ruBERT + AdaLoRA	0.89	0.92
ruBERT-Toxic (VK)	0.922	–

studies showing the advantages of contextual language representations for text classification.

The Kaggle DeepPavlov fine-tuning solution improved the Toxic F1 score to 0.83, confirming that pre-trained language models substantially outperform classical approaches on this dataset.

Among the models trained in this work, ruBERT-tiny achieved a Toxic F1 score of 0.88 and a Macro F1 score of 0.91. Despite its compact architecture, the model demonstrated strong classification performance and significantly outperformed the TF-IDF baseline.

The LoRA-based adaptation of DeepPavlov ruBERT achieved a Toxic F1 score of 0.87. Although the result was slightly below ruBERT-tiny, LoRA required training only a small fraction of the model parameters. This demonstrates that parameter-efficient fine-tuning can achieve competitive performance while substantially reducing computational costs.

The best result among the proposed approaches was obtained using AdaLoRA. The model achieved a Toxic F1 score of 0.89 and a Macro F1 score of 0.92. The adaptive rank allocation strategy provided a consistent improvement over standard LoRA and produced the strongest overall results among all models trained in this work.

The highest reported result on this dataset remains the ruBERT-Toxic model proposed by VK, which achieved a Toxic F1 score of 0.922. Although the proposed AdaLoRA approach did not surpass this result, the performance gap is relatively small. Considering that AdaLoRA updates only a subset of model parameters instead of performing full fine-tuning, the obtained results demonstrate the effectiveness of parameter-efficient adaptation techniques for Russian toxic comment classification.

Overall, the experiments show that modern transformer-based approaches significantly outperform classical machine learning methods. Furthermore, parameter-efficient fine-tuning methods such as LoRA and AdaLoRA are capable of achieving strong performance while requiring substantially fewer trainable parameters and shorter training times than full model fine-tuning.

7 Conclusion

In this work, the task of toxic comment classification in Russian was investigated using both classical machine learning and modern transformer-based approaches. A publicly available Russian toxic comments dataset was analyzed and divided into training, validation, and test subsets using stratified sampling.

Several models were evaluated, including a TF-IDF + Logistic Regression baseline, the compact ruBERT-tiny transformer model, and the DeepPavlov ruBERT-base model adapted using parameter-efficient fine-tuning techniques. In particular, LoRA and AdaLoRA were explored as computationally efficient alternatives to full model fine-tuning.

The experimental results demonstrated that transformer-based models significantly outperform classical machine learning approaches on the toxic comment classification task. Among the evaluated models, AdaLoRA achieved the best performance with a Toxic F1 score of 0.89 and a Macro F1 score of 0.92. Although the obtained result did not surpass the previously published state-of-the-art ruBERT-Toxic model, it achieved competitive performance while updating only a small fraction of the model parameters.

The experiments confirm that parameter-efficient fine-tuning methods can provide a strong balance between computational efficiency and classification quality. This makes such approaches particularly attractive in scenarios where hardware resources or training time are limited.

Future work may include experiments with larger language models, alternative PEFT methods, additional hyperparameter optimization, and evaluation on other Russian toxicity detection datasets.

References

- [1] Kaggle. *Russian Language Toxic Comments Dataset*. Available at: <https://www.kaggle.com/datasets/blackmoon/russian-language-toxic-comments/data> (Accessed: 30.05.2026).
- [2] Sismetanin. *Toxic Comments Detection in Russian*. GitHub repository. Available at: <https://github.com/sismetanin/toxic-comments-detection-in-russian> (Accessed: 30.05.2026).
- [3] Sedirov07. *Russian Text Classification with RuBERT*. Kaggle Notebook. Available at: <https://www.kaggle.com/code/sedirov07/russian-text-classification> (Accessed: 30.05.2026).
- [4] Westoon. *Toxic Comments Classifications RU*. Kaggle Notebook. Available at: <https://www.kaggle.com/code/westoon/toxic-comments-classifications-ru> (Accessed: 30.05.2026).
- [5] Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W. *LoRA: Low-Rank Adaptation of Large Language Models*. arXiv

preprint arXiv:2106.09685, 2021. Available at: <https://arxiv.org/abs/2106.09685>